Python Regular Expressions Cheat Sheet + Problem 1 Help

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1. Core Character Classes

Pattern	Matches	Example
`.`	Any single character (except newline)	`re.findall(r'.', 'cat')` → `['c','a','t']`
`\w`	Word char: letters, digits, underscore	`re.findall(r'\w', 'Hi_5!')` → `['H','i','_','5']`
`\W`	**Not** a word char	`re.findall(r'\W', 'Hi_5!')` → `['!']`
`\d`	Digit (0-9)	`re.findall(r'\d', 'Room 237')` → `['2','3','7']`
`\D`	**Not** a digit	`re.findall(r'\D', '123abc')` → `['a','b','c']`
`\s`	Whitespace (space, tab, newline)	`re.findall(r'\s', 'a b')` → `[' ']`
`\S`	**Not** whitespace	`re.findall(r'\S', 'a b')` → `['a','b']`

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2. Quantifiers (How Many Times?)

Pattern	Meaning	Example
`*`	0 or more times	`re.findall(r'\d*', 'a1b22')` → `['', '1', '', '22', '']`
`+`	1 or more times	`re.findall(r'\d+', 'a1b22')` → `['1', '22']`
`?`	0 or 1 time	`re.findall(r'colou?r', 'color colour')` → `['color', 'colour']`
`{n}`	Exactly n times	`re.findall(r'\d{3}', 'abc12345')` → `['123']`
`{m,n}`	Between m and n times	`re.findall(r'\d{2,4}', '12345')` → `['1234']`

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3. Anchors (Position in String)

Pattern	Matches	Example
`^`	Start of string	`re.findall(r'^Hi', 'Hi there')` → `['Hi']`
`$`	End of string	`re.findall(r'!$', 'Hello!')` → `['!']`
`\b`	Word boundary	`re.findall(r'\bcat\b', 'black cat')` → `['cat']`
`\B`	Not a word boundary	`re.findall(r'\Bcat', 'concatenate')` → `['cat']`

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4. Brackets and Groups

Pattern	Meaning	Example		
`[abc]`	`a` or `b` or `c`	`re.findall(r'[aeiou]', 'Hello')` → `['e','o']`		
`[^abc]`	NOT `a`, `b`, or `c`	`re.findall(r'[^aeiou]', 'Hello')` → consonants		
`(abc)`	Capture group	`re.findall(r'(ab)+', 'ababx')` → `['ab','ab']`		
`a	b`	`a` or `b`	`re.findall(r'cat	dog', 'catdog') → ['cat','dog']`

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5. Common Functions

Function	Purpose	Example
`re.search()`	Find first match anywhere	`re.search(r'\d+', 'ID 123')`
`re.match()`	Match from start	`re.match(r'ID', 'ID 123')`
`re.findall()`	Return all matches in list	`re.findall(r'\w+', 'Hello world!')` → `['Hello','world']`
`re.sub()`	Replace matches	`re.sub(r'\d', '#', 'Call 911')` → `'Call ###'`
`re.split()`	Split on matches	`re.split(r'\s+', 'a   b c')` → `['a','b','c']`

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6. Problem 1 – Movie Quote Normalizer (Step-by-Step Help)

Goal: Lowercase, remove punctuation, collapse spaces, split into words. Raise `ValueError` if input is empty/whitespace.

Code:

```
import re

def normalize_quote(text: str) -> list[str]:
    if not isinstance(text, str):
        raise TypeError("text must be a str")
    if not text.strip():
        raise ValueError("text is empty/whitespace")

    text = text.lower()
    text = re.sub(r"[^\w\s]", " ", text)  # remove punctuation
    words = text.split()
    return words

# Example
quote = "  May  THE  Force—be with you!! "
print(normalize_quote(quote))  # ['may', 'the', 'force', 'be', 'with', 'you']
```